

Data Logging System for Physiological Parameters

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Outline

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Future Work



- ▶ Continuous monitoring of physiological parameters like heart rate, ECG, temperature is essential for early detection of health issues.
- ▶ Traditional healthcare systems rely on manual and periodic measurements, which may miss critical real-time changes.
- ▶ With advancements in embedded systems and IoT, real-time data acquisition and monitoring have become more efficient and accessible.
- ▶ This project focuses on developing a data logging system that collects, processes, and stores physiological data continuously.
- ▶ The system also includes an intelligent alert mechanism to notify abnormal conditions in real-time.



Problem Statement

- ▶ Continuous monitoring of physiological parameters is essential for effective patient care.
- ▶ Existing systems often rely on manual monitoring, which is time-consuming and prone to human error.
- ▶ Sudden abnormal changes in patient parameters may go undetected in real-time.
- ▶ Delay in identifying critical conditions can lead to serious health risks.
- ▶ There is a need for an automated system to continuously monitor, acquire, and analyze physiological data.
- ▶ The system should also provide real-time alerts for abnormal parameter values.



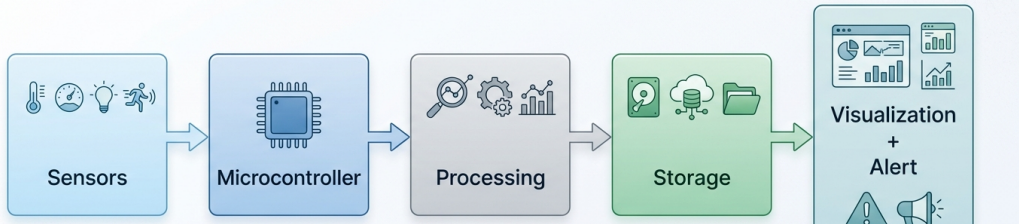
Objectives

- ▶ To acquire and log real-time data efficiently using the TM4C123GXL microcontroller for further analysis.
- ▶ To design a system for continuous monitoring of physiological parameters such as heart rate, ECG, and temperature.
- ▶ To develop a mechanism for processing and analyzing physiological data.
- ▶ To implement an intelligent alert system for detecting abnormal conditions.
- ▶ To ensure reliable storage and retrieval of patient data.
- ▶ To create a user-friendly interface for visualization of physiological parameters.



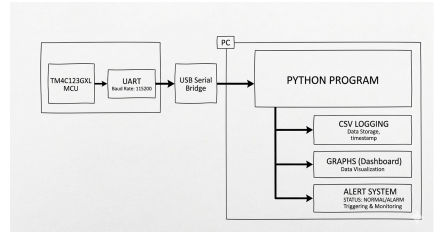
System Overview

- ▶ The proposed system is a real-time data logging system for monitoring physiological parameters such as heart rate, ECG, and temperature.
- ▶ Sensors are used to acquire physiological signals from the patient, which are then retrieved using a microcontroller.
- ▶ The acquired data is transmitted to a system for storage and further analysis.
- ▶ The system continuously logs data, enabling tracking of patient health over time.
- ▶ An alert mechanism is incorporated to detect abnormal parameter values and notify users in real-time. The collected data is visualized through graphs or dashboards for easy interpretation.



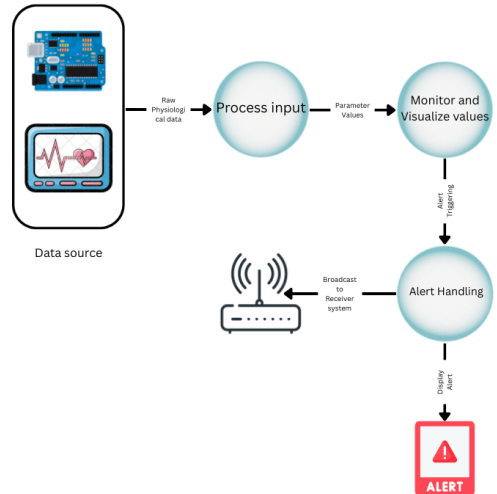
System Architecture

- ▶ TM4C123GXL microcontroller generates simulated physiological parameters.
- ▶ Data is transmitted using UART serial communication.
- ▶ The UART data is received by the computer through the USB serial interface.
- ▶ A Python monitoring program reads the incoming data through the COM port using PySerial.
- ▶ The received data is stored in a CSV file.
- ▶ Matplotlib generates graphs for visualization of physiological parameters.
- ▶ The system detects abnormal values and generates alerts.



System Workflow

- ▶ Data Acquisition using TM4C123GXL
- ▶ Transmission of physiological data
- ▶ Data reception by monitoring system
- ▶ Data processing and monitoring
- ▶ Graphical visualization of parameters
- ▶ Detection of abnormal values
- ▶ Alert generation
- ▶ Broadcast alerts to receiver systems



Software Tools Used

- ▶ **Code Composer Studio (CCS)** Used to write, compile and upload embedded C programs to the TM4C123GXL microcontroller.
- ▶ **Python** Used to receive and process the physiological data sent by the microcontroller.
- ▶ **Pandas** Python library used for storing and handling physiological data in structured format.
- ▶ **Matplotlib** Used to plot graphs for visualization of physiological parameters over time.
- ▶ **PySerial** Python library used for serial communication between TM4C123GXL and the monitoring system.



Implementation Methodology

- ▶ The system is implemented using a microcontroller interfaced with a physiological sensor and a host system.
- ▶ The microcontroller acquires real-time data from sensor using UART communication.
- ▶ The system operates in two modes: Real Sensor Mode and Simulation Mode, which can be toggled using a push button.
- ▶ In Simulation Mode, physiological parameters are generated programmatically, including both normal and abnormal values.
- ▶ The acquired or simulated data is transmitted to the system via UART for further processing and visualization.
- ▶ LED indicators are used to represent system states such as data acquisition, waiting state, and simulation mode.
- ▶ Further processing, analysis, and alert generation are performed at the system level.



Physiological Parameters Monitored

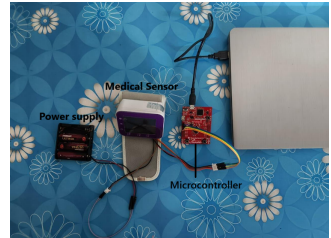
- ▶ Heart Rate (HR)
- ▶ Oxygen Saturation (SpO2)
- ▶ Respiratory Rate (RR)
- ▶ Body Temperature
- ▶ Blood Pressure (BP)

	Timestamp	PatientID	HR	BP	SpO2	Temp	RR
1	20260311192813	P12345	118	171/106	83	38.5	27
2	20260311192816	P12345	94	110/80	99	36.4	12
3	20260311192819	P12345	75	117/83	99	36.7	16
4	20260311192822	P12345	86	123/75	96	36.9	12
5	20260311192825	P12345	92	116/74	98	36.7	15
6	20260311192828	P12345	86	110/74	100	37.0	16
7	20260311192831	P12345	65	170/69	82	36.6	21
8	20260311192834	P12345	81	124/73	99	36.9	16
9	20260311192837	P12345	84	114/74	96	36.6	15
10	20260311192840	P12345	78	121/74	96	36.4	13
11	20260311192843	P12345	67	122/77	100	36.7	12
12	20260311192846	P12345	82	112/83	100	36.7	14
13	20260311192849	P12345	69	112/83	99	37.0	18
14	20260311192852	P12345	86	111/78	96	36.5	16
15	20260311192855	P12345	65	114/80	99	36.8	13
16	20260311192858	P12345	70	125/76	97	36.4	14
17	20260311192901	P12345	59	111/80	91	39.0	23
18	20260311192904	P12345	72	124/84	99	36.7	14
19	20260311192907	P12345	83	112/78	98	37.1	12



Hardware Setup

- ▶ The system consists of physiological sensing device(s), a microcontroller unit, power supply, and communication interface.
- ▶ The sensing device is used to capture physiological parameters such as heart rate and oxygen saturation.
- ▶ The microcontroller is interfaced with the sensor to acquire real-time data.
- ▶ A battery module is used to provide power to the sensing device and supporting components.
- ▶ The microcontroller is connected to the system using a communication interface (USB/UART) for data transmission.
- ▶ Proper wiring and connections ensure reliable data acquisition and communication.



Data Acquisition and Transmission

- ▶ Physiological data is acquired from the blood pressure sensor using UART communication.
- ▶ The microcontroller continuously listens for incoming sensor data in a non-blocking manner.
- ▶ Received data packets are parsed to extract systolic, diastolic, and pulse values.
- ▶ The system operates in two modes: Real Sensor Mode and Simulation Mode.
- ▶ In Simulation Mode, physiological parameters are generated programmatically for testing purposes.
- ▶ The acquired or simulated data is formatted and transmitted to the host system via UART.
- ▶ The transmitted data is used for further processing, visualization, and alert generation.



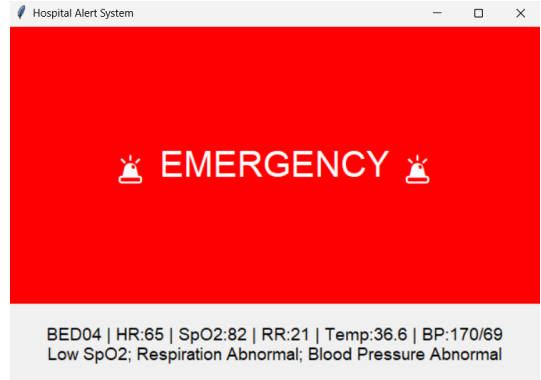
Graph Plotting

- ▶ The system provides real-time visualization of physiological parameters using graphical plots.
- ▶ Parameters displayed include Heart Rate, SpO2, Respiration Rate, Temperature, and Blood Pressure.
- ▶ Each parameter is plotted against time to observe trends and variations.
- ▶ The visualization helps in identifying abnormal patterns in physiological data.
- ▶ Both normal and abnormal conditions can be clearly distinguished through graphical changes.



Alert Generation System

- ▶ The system continuously monitors incoming data.
- ▶ Each parameter is compared with predefined normal ranges.
- ▶ If abnormal values are detected, alerts are generated.
- ▶ Alerts are broad-casted to all systems running the receiver program.



AI-Based Features

AI Summary (Clinical Analysis)

- ▶ The system analyzes real-time physiological data such as Heart Rate, SpO2, Temperature, and Blood Pressure. Statistical measures (average, minimum, maximum) are computed to identify trends.
- ▶ AI generates a structured clinical report indicating patient status (e.g., Stable or Critical). Provides concise insights on abnormalities and physiological correlations.

AI Voice Assistant

- ▶ Enables users to interact with the system using voice commands. Speech input is converted to text and processed using AI models.
- ▶ The system retrieves relevant patient data from logs based on the query. AI-generated responses are converted into speech for user-friendly interaction.

Clinical Insights

ANALYSIS REPORT

****STATUS: STABLE (Based on Available Data)****

****SUMMARY OF TRENDS:****
The patient's minimum SpO2 of 98% and maximum temperature of 36.5°C indicate good oxygenation and normal thermoregulation, respectively. However, a comprehensive trend analysis is severely limited by the absence of heart rate, blood pressure, and the latest SpO2 values. Therefore, while the available data is reassuring, a complete and accurate clinical picture requires the collection of these critical missing vital signs.

****PHYSIOLOGICAL CORRELATIONS BETWEEN SpO2 AND HR:****
In a typical physiological response, there is an inverse relationship between SpO2 and HR when hypoxemia is present. If SpO2 (oxygen saturation) drops, the body often attempts to compensate by increasing the heart rate (tachycardia) to enhance oxygen delivery to the tissues. Conversely, stable and optimal SpO2 values are generally associated with a normal, stable heart rate in the absence of other stressors. However, given that both the average heart rate and the latest heart rate, as well as the latest SpO2, are listed as 'nan' in the provided data, no specific physiological correlation or trend between these two vital signs can be identified or analyzed for this particular patient at this time.

Dismiss

```
AI: I am ready. Which time should I check in your logs?
Listening...
Processing voice...
You: check for 13 hours 51 minutes 29 seconds
AI: Based on the provided health logs, I found the closest match for 13 hours 51 minutes 29 seconds. The data for 13:54:29 is as follows:
Heart Rate (HR): 70.0
SpO2: 98.0
Temperature (Temp): 36.4

Since HR (70.0) is within the normal range and Temp (36.4) is also within the normal range, there is no health spike warning.
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Work Completed in Phase 2

- ▶ Implementation of AI-based clinical summary for analyzing physiological data.
- ▶ Development of a voice assistant for interactive, hands-free data access.
- ▶ Integration of real-time medical sensor data retrieval from the system.
- ▶ Enhancement of data processing and visualization modules.
- ▶ Improvement in system reliability and user interaction.



- ▶ Integration of additional physiological sensors for comprehensive health monitoring.
- ▶ Implementation of advanced AI/ML models for accurate prediction and diagnosis.
- ▶ Cloud-based data storage for remote access and long-term analysis.
- ▶ Mobile application development for real-time monitoring and alerts.
- ▶ Enhancement of alert system with automated emergency notifications.
- ▶ Improvement in system scalability and deployment in real healthcare environments.



Conclusion

- ▶ A real-time physiological data logging and monitoring system was successfully developed.
- ▶ The system enables continuous acquisition, transmission, and visualization of patient data.
- ▶ AI-based analysis provides meaningful clinical insights from raw data.
- ▶ The voice assistant enhances user interaction and accessibility.
- ▶ The system demonstrates potential for efficient and intelligent healthcare monitoring.



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Many Thanks

